

April 14th Quiz

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Econ 1007

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(1) Given $U(x,y) = 12x^{0.5}y^{0.5}$

(where good x & good y are measured in "units" AND $U(x,y)$ is measured in "utils")

If $U(x,y) = 96$ (isolate of the constant equation)

(a) Get the indifference curve equation: $y(x)$:

$$96 = 12x^{0.5}y^{0.5}$$

$$y = \left(\frac{96}{12}\right)x^{-0.5}$$

$$y = \left(\frac{96}{12}\right)x^{-0.5} \text{ or indifference curve}$$

(b) Get the MRS $\left(\frac{dy}{dx}\right)$ for this indifference curve:

$$\frac{dy}{dx} = \left(\frac{96}{12}\right)(-1)x^{-1.5} = \left(\frac{-8}{x^{1.5}}\right) = \text{MRS}$$

(c) Calculate the MRS when $x = 2$:

$$\text{MRS}_{(x=2)} = \frac{dy}{dx} = \frac{-8}{x^{1.5}} = \frac{-8}{4} = -2 < 0$$

(d) Interpret your result from question (3):

So, indifference curve has a Diminishing Marginal Rate of Substitution

When consuming 2 units of good X (and thus $y = \left(\frac{96}{12x}\right) = \left(\frac{96}{12(2)}\right) = \left(\frac{8}{2}\right) = 4$ units of good Y), if the consumer buys 1 more unit of good X, holding all other things equal (including overall utility in utils)

his/her consumption of good Y must decrease by 2 units. QED

*NB, if $x=7, y=1.143 \Rightarrow \frac{dy}{dx} = \text{MRS} = \frac{-8}{x^{1.5}} = -0.163$ Diminishing Marginal Rate of Substitution.